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Comparative Analysis of Clinical and Imaging Characteristics Between Mycobacterium avium Complex (MAC) and Mycobacterium abscessus Complex (MABC) in Non-Tuberculous Mycobacterial Pulmonary Disease: A Retrospective Single-Centre Study from Southern China

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Abstract

Background: The objective of this study is to undertake a comprehensive comparison of the clinical and imaging characteristics exhibited by pulmonary diseases caused by Mycobacterium avium complex (MAC) and Mycobacterium abscessus complex (MABC). The findings of this study have the potential to provide insights that could refine diagnostic approaches and therapeutic interventions for these challenging infections.

Methods: A retrospective study of 134 patients (90 MAC and 44 MABC) with pathogen-confirmed non-tuberculous mycobacterial pulmonary disease was conducted at the First Affiliated Hospital of Guangxi Medical University. Univariate

analysis (Mann-Whitney U, t-tests, Chi-Square) and multivariate logistic regression were performed to identify discriminative factors.

Results: The present study has demonstrated a significant association between the Zhuang ethnicity and MABC infection (59.1% vs. 31.1% in MAC, $p=0.002$). Furthermore, the study found that MABC patients tended to be younger, with a mean age of 54.9 years compared to 58.7 years in the control group, although this difference did not achieve statistical significance ($p=0.098$). No statistically significant differences were observed with respect to sex, BMI, or occupational distribution ($p > 0.05$) (Table 1). The prevalence of constitutional symptoms was also observed to be similar in the MAC and MABC cohorts. Laboratory analyzes indicated comparable immunological and inflammatory profiles. A comparison of the two cohorts revealed distinct dissemination patterns for MABC and MAC infections. Furthermore, multivariate logistic regression analysis revealed several significant predictors. These include an inverse association between age and MAC pulmonary disease (OR = 0.97/year, 95% CI: 0.95-0.99, $p = 0.015$), a strong association between bone involvement and MAC (OR = 3.89, 95% CI: 1.13-13.36, $p = 0.031$) and a reduced risk of MABC associated with lymph node involvement (OR = 0.26, 95% CI: 0.10-0.68, $p = 0.006$). The occurrence of a *Talaromyces marneffei* co-infection was found to result in a threefold increase in the risk of MAC (odds ratio [OR] = 3.20, 95% confidence interval [CI] = 1.08-9.47, $p = 0.036$).

Conclusions: This study provides a comprehensive comparison of clinical and imaging characteristics between lung diseases of MAC and MABC, identifying several potential predictors that can aid in differential diagnosis. These findings underscore the different pathogenic behaviors and host interactions of MAC and MABC, supporting targeted diagnostic approaches and therapeutic strategies. Future multicenter studies with larger cohorts are warranted to validate these results and further explore the biological mechanisms underlying the identified predictors.

Keywords: *Mycobacterium avium* complex (MAC), *Mycobacterium abscessus* complex (MABC), non-tuberculous mycobacterial pulmonary disease, clinical characteristics, imaging characteristics, retrospective study, single-center study

1 Introduction

The Introduction section, of referenced text [1] expands on the background of the work (some overlap with the Abstract is acceptable). The introduction should not include subheadings.

Springer Nature does not impose a strict layout as standard however authors are advised to check the individual requirements for the journal they are planning to submit to as there may be journal-level preferences. When preparing your text please also be aware that some stylistic choices are not supported in full text XML (publication version), including coloured font. These will not be replicated in the typeset article if it is accepted.

2 Methods

2.1 Study Design and Population

This retrospective cohort study was conducted at the First Affiliated Hospital of Guangxi Medical University, China. We systematically reviewed medical records of 282 patients diagnosed with non-tuberculous mycobacterial pulmonary disease (NTM-PD) between January 2020 and December 2024. Diagnosis was confirmed via bronchoalveolar lavage fluid (BALF) culture and metagenomic next-generation sequencing (mNGS), strictly adhering to the 2020 American Thoracic Society (ATS) diagnostic guidelines. After applying exclusion criteria—mixed infections, incomplete clinical records, non-MAC/MABC species, HIV co-infection, and colonization cases (as defined by ATS criteria)—134 patients were included in the final analysis: 90 MAC and 44 MABC cases.

2.2 Data Collection and Variables

Standardized case report forms were used to extract multidimensional data: 1. **Demographics:** Age, sex, BMI, occupation (agricultural vs. non-agricultural), ethnicity (Han Chinese vs. Zhuang); Exposure history: smoking status, alcohol use, and occupational dust exposure. 2. **Comorbidities and Pharmacological history:** prior tuberculosis, COPD, bronchiectasis, type 2 diabetes, malignancies, autoimmune diseases (including anti-IFN- γ autoantibody syndrome), anemia, and peptic ulcer disease and chronic corticosteroid use (>3 months). 3. **Clinical Presentation:** fever, night sweats, weight loss, fatigue, chronic cough, sputum production, hemoptysis, dyspnea, chest pain, abdominal distension, and poor appetite. 4. **Laboratory Investigations:** leukocyte count, neutrophil-to-lymphocyte ratio (NLR), procalcitonin (PCT), IL-6, CD4+/CD8+ T-cell counts, CD4/CD8 ratio, and serum albumin. 5. **Radio-logical Assessment (High-resolution CT):** bronchiectasis, cavitation, nodules, consolidation, effusion, thickening, and Lymphatic spread (mediastinal/axillary lymphadenopathy). 6. **Extrapulmonary dissemination:** bone, skin, or multiorgan involvement.

2.3 Statistical Analysis

Univariate Analysis: - **Continuous variables:** Mann-Whitney U test (non-normal distribution) or independent t-test (normal distribution, assessed via Shapiro-Wilk test). - **Categorical variables:** Chi-Square or Fisher's exact test (expected cell counts < 5).

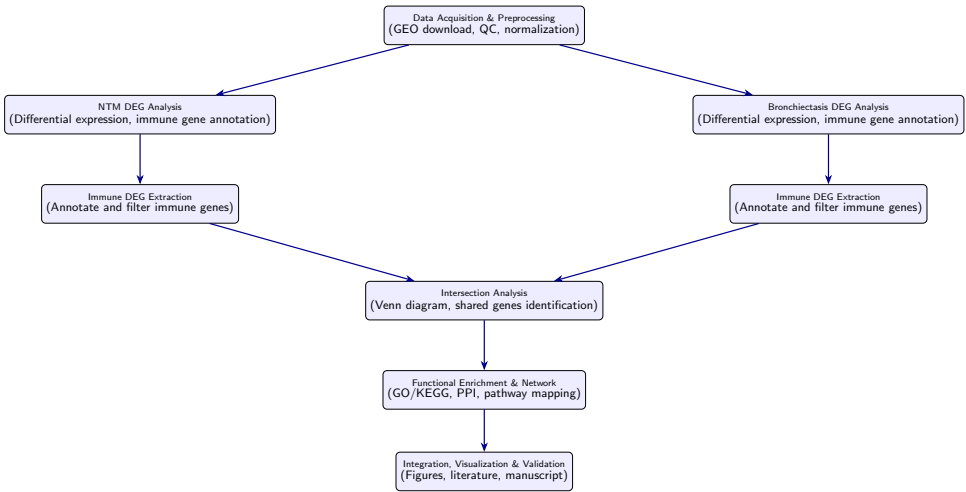
Multivariable Modeling: - Variables with $p \leq 0.1$ in univariate analysis and clinically relevant factors (age, BMI) were entered into a backward stepwise logistic regression ($P_{in}=0.05$, $P_{out}=0.10$). - **Model performance** was evaluated using: - Hosmer-Lemeshow goodness-of-fit test - Area under the ROC curve (AUC) with 95% confidence intervals - Nagelkerke R^2 for explained variance

Software: SPSS 27.0 (IBM) and R 4.3.1 (glmnet, mice packages).

2.4 Study flowchart

The study workflow is illustrated in Figure 3, summarizing the stepwise bioinformatic analysis from raw data acquisition to integrative interpretation.

Fig. 1. Bioinformatic workflow for identifying shared immunological mechanisms between NTM and bronchiectasis. Each box represents a key analytical step; arrows indicate process flow.



Study Flowchart2
Study Flowchat3

2.5 Ethical Considerations

The study protocol was approved by the Institutional Review Board of Guangxi Medical University. Written informed consent was waived due to the retrospective nature of the study, adhering to national regulations and the Declaration of Helsinki.

3 Results

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Fig. 2. Bioinformatic workflow for identifying shared immunological mechanisms between NTM and bronchiectasis. Each box represents a key analytical step; arrows indicate process flow.

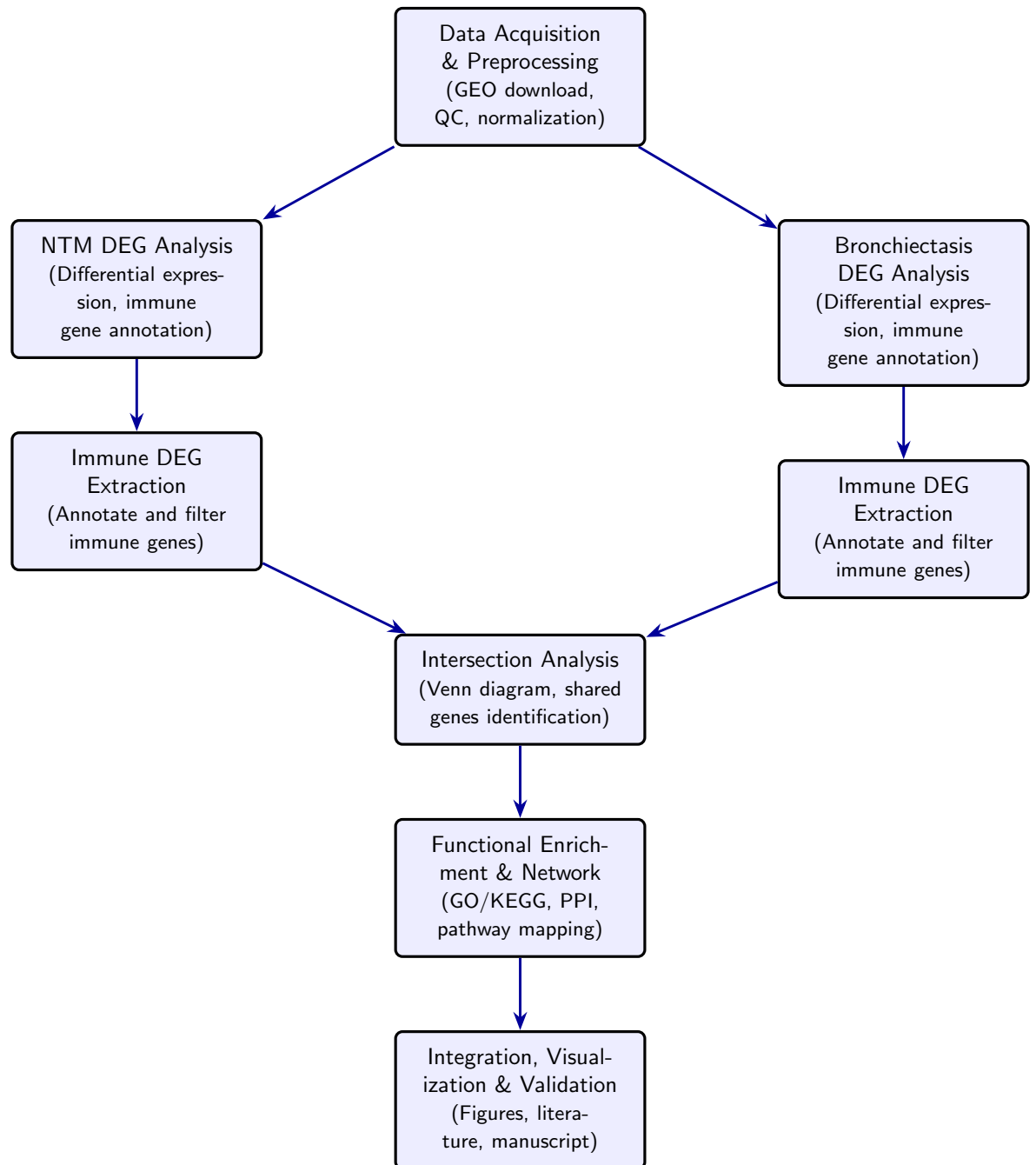


Fig. 3. Bioinformatic workflow for identifying shared immunological mechanisms between NTM and bronchiectasis. Each box represents a key analytical step; arrows indicate process flow.

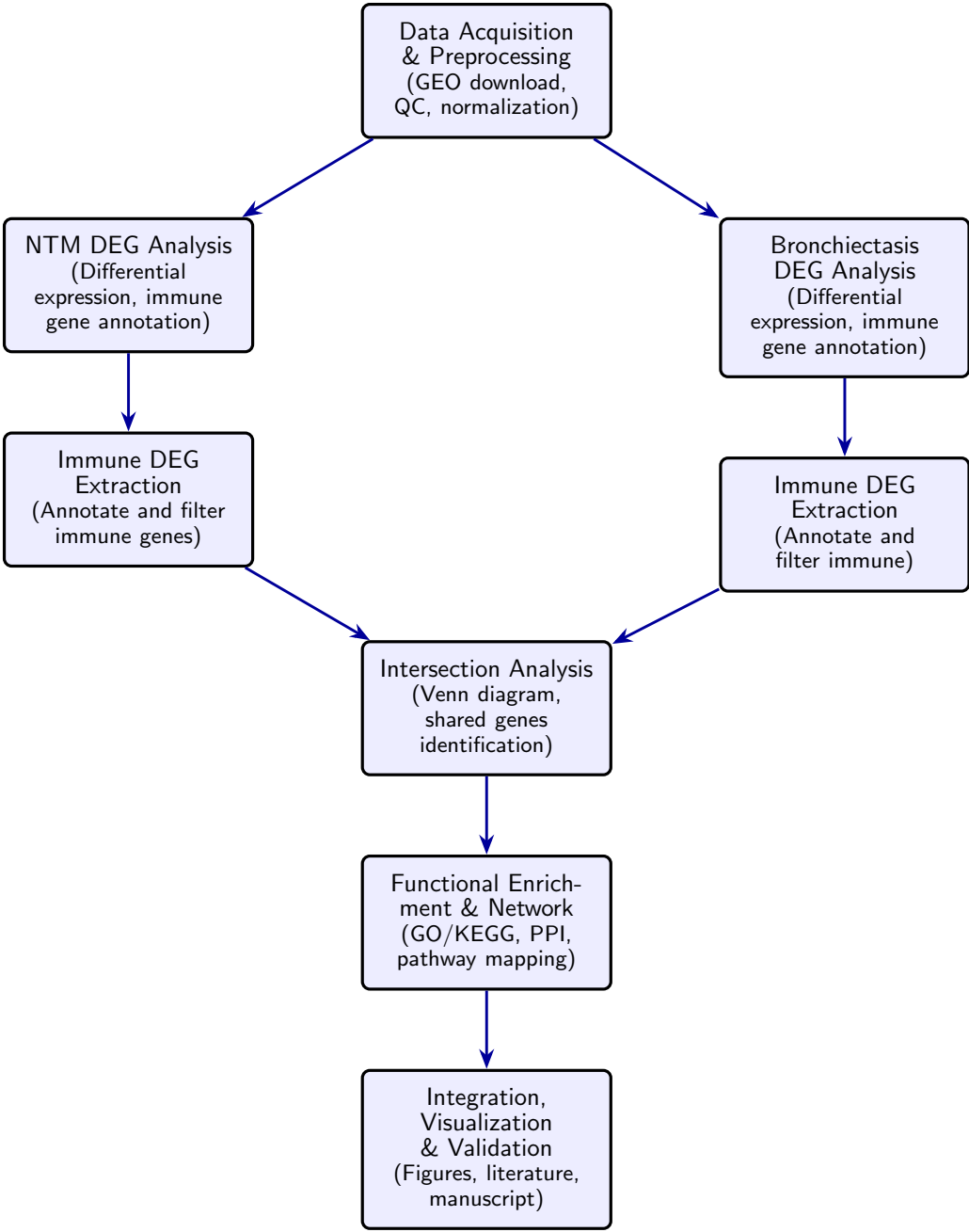


Table 1: Baseline demographic and clinical characteristics of MAC and MABC patients

Characteristic	MAC (n=90)	MABC (n=44)
Age, years, mean (SD)	58.7 (13.0)	54.9 (11.8)
Male, n (%)	46 (51.1)	23 (52.3)
BMI, kg/m ² , mean (SD)	21.4 (3.2)	21.0 (3.5)
Zhuang ethnicity, n (%)	28 (31.1)	26 (59.1)**
Agricultural occupation, n (%)	35 (38.9)	18 (40.9)
Smoking history, n (%)	22 (24.4)	11 (25.0)
Alcohol use, n (%)	14 (15.6)	8 (18.2)
COPD, n (%)	10 (11.1)	4 (9.1)
Bronchiectasis, n (%)	40 (44.4)	21 (47.7)
Diabetes, n (%)	12 (13.3)	6 (13.6)

Note: Data are shown as mean (SD) or n (%). ** $p < 0.01$ vs. MAC.

Table 2: Univariate analysis of clinical and laboratory variables associated with MAC and MABC

Variable	MAC (n=90)	MABC (n=44)	<i>p</i> -value
Age, years, mean (SD)	58.7 (13.0)	54.9 (11.8)	0.098
Male, n (%)	46 (51.1)	23 (52.3)	0.891
Zhuang ethnicity, n (%)	28 (31.1)	26 (59.1)	0.002
Fever, n (%)	33 (36.7)	18 (40.9)	0.641
Night sweats, n (%)	10 (11.1)	4 (9.1)	0.726
Weight loss, n (%)	21 (23.3)	12 (27.3)	0.622
Albumin, g/L, mean (SD)	38.2 (5.0)	37.8 (5.6)	0.668
NLR, median (IQR)	3.2 (2.2–4.5)	3.4 (2.4–4.7)	0.550
Bronchiectasis, n (%)	40 (44.4)	21 (47.7)	0.752
Cavitation, n (%)	26 (28.9)	8 (18.2)	0.163
Bone involvement, n (%)	7 (7.8)	1 (2.3)	0.280
Lymphadenopathy, n (%)	14 (15.6)	15 (34.1)	0.019
Talaromyces marneffei co-infection	10 (11.1)	1 (2.3)	0.104

Note: NLR = neutrophil-to-lymphocyte ratio; IQR = interquartile range.

Table 3: Multivariate logistic regression for predictors of MABC infection (vs. MAC)

Variable	Odds Ratio (OR)	95% CI	<i>p</i> -value
Age (per year increase)	0.97	0.95–0.99	0.015
Zhuang ethnicity	2.89	1.32–6.31	0.008
Bone involvement	3.89	1.13–13.36	0.031
Lymph node involvement	0.26	0.10–0.68	0.006
T. marneffei co-infection	3.20	1.08–9.47	0.036

Note: CI = confidence interval; reference group is MAC.

Table 4: Multivariate logistic regression for predictors of MABC infection (vs. MAC)

Variable	Odds Ratio (OR)	95% CI	<i>p</i> -value
Age (per year increase)	0.97	0.95–0.99	0.015
Zhuang ethnicity	2.89	1.32–6.31	0.008
Bone involvement	3.89	1.13–13.36	0.031
Lymph node involvement	0.26	0.10–0.68	0.006
T. marneffeii co-infection	3.20	1.08–9.47	0.036

Note: CI = confidence interval; reference group is MAC.

- 3.1 Demographic Characteristics
- 3.2 Comorbidities and Underlying Conditions
- 3.3 Clinical Symptoms and Laboratory Profiles
- 3.4 Radiological Phenotypes and Disease Dissemination
- 3.5 Multivariable Predictors of MAC vs. MABC Pulmonary Disease

Table 5: Multivariate logistic regression for predictors of MABC infection (vs. MAC)

Variable	Odds Ratio (OR)	95% CI	<i>p</i> -value
Age (per year increase)	0.97	0.95–0.99	0.015
Zhuang ethnicity	2.89	1.32–6.31	0.008
Bone involvement	3.89	1.13–13.36	0.031
Lymph node involvement	0.26	0.10–0.68	0.006
T. marneffeii co-infection	3.20	1.08–9.47	0.036

Note: CI = confidence interval; reference group is MAC.

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 - 4.1.1 This is an example for third level head—subsubsection head

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Table 6: Multivariate logistic regression for predictors of MABC infection (vs. MAC)

Variable	Odds Ratio (OR)	95% CI	<i>p</i> -value
Age (per year increase)	0.97	0.95–0.99	0.015
Zhuang ethnicity	2.89	1.32–6.31	0.008
Bone involvement	3.89	1.13–13.36	0.031
Lymph node involvement	0.26	0.10–0.68	0.006
T. marneffeii co-infection	3.20	1.08–9.47	0.036

Note: CI = confidence interval; reference group is MAC.

Table 7: Multivariate logistic regression for predictors of MABC infection (vs. MAC)

Variable	Odds Ratio (OR)	95% CI	<i>p</i> -value
Age (per year increase)	0.97	0.95–0.99	0.015
Zhuang ethnicity	2.89	1.32–6.31	0.008
Bone involvement	3.89	1.13–13.36	0.031
Lymph node involvement	0.26	0.10–0.68	0.006
T. marneffeii co-infection	3.20	1.08–9.47	0.036

Note: CI = confidence interval; reference group is MAC.

Table 8: Baseline demographic and clinical characteristics of MAC and MABC patients

Characteristic	MAC (n=90)	MABC (n=44)
Age, years, mean (SD)	58.7 (13.0)	54.9 (11.8)
Male, n (%)	46 (51.1)	23 (52.3)
BMI, kg/m ² , mean (SD)	21.4 (3.2)	21.0 (3.5)
Zhuang ethnicity, n (%)	28 (31.1)	26 (59.1)**
Agricultural occupation, n (%)	35 (38.9)	18 (40.9)
Smoking history, n (%)	22 (24.4)	11 (25.0)
Alcohol use, n (%)	14 (15.6)	8 (18.2)
COPD, n (%)	10 (11.1)	4 (9.1)
Bronchiectasis, n (%)	40 (44.4)	21 (47.7)
Diabetes, n (%)	12 (13.3)	6 (13.6)

Note: Data are shown as mean (SD) or n (%). ***p* < 0.01 vs. MAC.

5 Equations

Equations in L^AT_EX can either be inline or on-a-line by itself (“display equations”). For inline equations use the $\dots$ commands. E.g.: The equation $H\psi = E\psi$ is written via the command $\text{\$H \psi = E \psi\$}$.

Table 9: Univariate analysis of clinical and laboratory variables associated with MAC and MABC

Variable	MAC (n=90)	MABC (n=44)	p-value
Age, years, mean (SD)	58.7 (13.0)	54.9 (11.8)	0.098
Male, n (%)	46 (51.1)	23 (52.3)	0.891
Zhuang ethnicity, n (%)	28 (31.1)	26 (59.1)	0.002
Fever, n (%)	33 (36.7)	18 (40.9)	0.641
Night sweats, n (%)	10 (11.1)	4 (9.1)	0.726
Weight loss, n (%)	21 (23.3)	12 (27.3)	0.622
Albumin, g/L, mean (SD)	38.2 (5.0)	37.8 (5.6)	0.668
NLR, median (IQR)	3.2 (2.2–4.5)	3.4 (2.4–4.7)	0.550
Bronchiectasis, n (%)	40 (44.4)	21 (47.7)	0.752
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Bone involvement, n (%)	7 (7.8)	1 (2.3)	0.280
Lymphadenopathy, n (%)	14 (15.6)	15 (34.1)	0.019
Talaromyces marneffei co-infection	10 (11.1)	1 (2.3)	0.104

Note: NLR = neutrophil-to-lymphocyte ratio; IQR = interquartile range.

Table 10: Multivariate logistic regression for predictors of MABC infection (vs. MAC)

Variable	Odds Ratio (OR)	95% CI	p-value
Age (per year increase)	0.97	0.95–0.99	0.015
Zhuang ethnicity	2.89	1.32–6.31	0.008
Bone involvement	3.89	1.13–13.36	0.031
Lymph node involvement	0.26	0.10–0.68	0.006
T. marneffei co-infection	3.20	1.08–9.47	0.036

Note: CI = confidence interval; reference group is MAC.

For display equations (with auto generated equation numbers) one can use the equation or align environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum_{i=1}^p \|\tilde{Y}_i(k)\|^2 + \sum_{j=1}^q \|\tilde{Z}_j(k)\|^2}{p+q}. \quad (1)$$

where,

$$\begin{aligned} D_\mu &= \partial_\mu - ig \frac{\lambda^a}{2} A_\mu^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^a \end{aligned} \quad (2)$$

Notice the use of `\nonumber` in the align environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation numbers are required. The `\label{}` command should only be used at the last line of an align

environment where `\nonumber` is not used.

$$Y_{\infty} = \left(\frac{m}{\text{GeV}}\right)^{-3} \left[1 + \frac{3 \ln(m/\text{GeV})}{15} + \frac{\ln(c_2/5)}{15}\right] \quad (3)$$

The class file also supports the use of `\mathbb{}`, `\mathscr{}` and `\mathcal{}` commands. As such `\mathbb{R}`, `\mathscr{R}` and `\mathcal{R}` produces $\mathbb{R}$, $\mathscr{R}$ and $\mathcal{R}$ respectively (refer Subsubsection 4.1.1).

6 Tables

Tables can be inserted via the normal table and tabular environment. To put footnotes inside tables you should use `\footnotetext[]{\dots}` tag. The footnote appears just below the table itself (refer Tables 11 and 12). For the corresponding footnotemark use `\footnotemark[\dots]`

Table 11: Caption text

Column 1	Column 2	Column 3	Column 4
row 1	data 1	data 2	data 3
row 2	data 4	data 5 ¹	data 6
row 3	data 7	data 8	data 9 ²

Source: This is an example of table footnote. This is an example of table footnote.

¹Example for a first table footnote. This is an example of table footnote.

²Example for a second table footnote. This is an example of table footnote.

The input format for the above table is as follows:

```
\begin{table}[<placement-specifier>]
\caption{<table-caption>}\label{<table-label>}%
\begin{tabular}{@{}l l l l@{}}
\toprule
Column 1 & Column 2 & Column 3 & Column 4\\
\midrule
row 1 & data 1 & data 2 & data 3 \\
row 2 & data 4 & data 5\footnotemark[1] & data 6 \\
row 3 & data 7 & data 8 & data 9\footnotemark[2]\\
\botrule
\end{tabular}
\footnotetext{Source: This is an example of table footnote.}
```



```

This is an example of table footnote.}
\footnotetext[1]{Example for a first table footnote.
This is an example of table footnote.}
\footnotetext[2]{Example for a second table footnote.
This is an example of table footnote.}
\end{table}

```

Table 12: Example of a lengthy table which is set to full textwidth

Project	Element 1 ¹			Element 2 ²		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Note: This is an example of table footnote. This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

¹Example for a first table footnote.

²Example for a second table footnote.

In case of double column layout, tables which do not fit in single column width should be set to full text width. For this, you need to use `\begin{table*}` ... `\end{table*}` instead of `\begin{table}` ... `\end{table}` environment. Lengthy tables which do not fit in textwidth should be set as rotated table. For this, you need to use `\begin{sidewaystable}` ... `\end{sidewaystable}` instead of `\begin{table*}` ... `\end{table*}` environment. This environment puts tables rotated to single column width. For tables rotated to double column width, use `\begin{sidewaystable*}` ... `\end{sidewaystable*}`.

7 Figures

As per the $\text{\LaTeX}$ standards you need to use eps images for $\text{\LaTeX}$ compilation and pdf/jpg/png images for PDF $\text{\LaTeX}$ compilation. This is one of the major difference between $\text{\LaTeX}$ and PDF $\text{\LaTeX}$. Each image should be from a single input .eps/vector image file. Avoid using subfigures. The command for inserting images for $\text{\LaTeX}$ and PDF $\text{\LaTeX}$ can be generalized. The package used to insert images in $\text{\LaTeX}$ /PDF $\text{\LaTeX}$ is the graphicx package. Figures can be inserted via the normal figure environment as shown in the below example:

```

\begin{figure}[<placement-specifier>]
\centering
\includegraphics{<eps-file>}

```

Table 13: Tables which are too long to fit, should be written using the “sidewaystable” environment as shown here

Projectile	Element 1 ¹				Element ²		
	Energy	σ_{calc}	σ_{expt}		Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 $\pm$ 12		780 A	1166	1239 $\pm$ 100
Element 4	500 A	961	922 $\pm$ 10		900 A	1268	1092 $\pm$ 40
Element 5	990 A	1168	1547 $\pm$ 12		780 A	1166	1239 $\pm$ 100
Element 6	500 A	961	922 $\pm$ 10		900 A	1268	1092 $\pm$ 40

Note: This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote
this is an example of table footnote.

¹This is an example of table footnote.

\caption{<figure-caption>}\label{<figure-label>}

\end{figure}

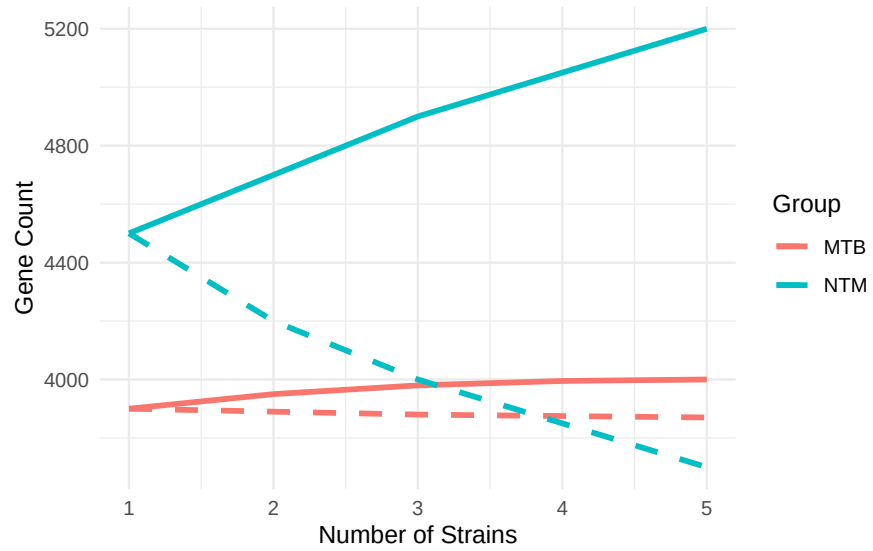


Fig. 4. This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

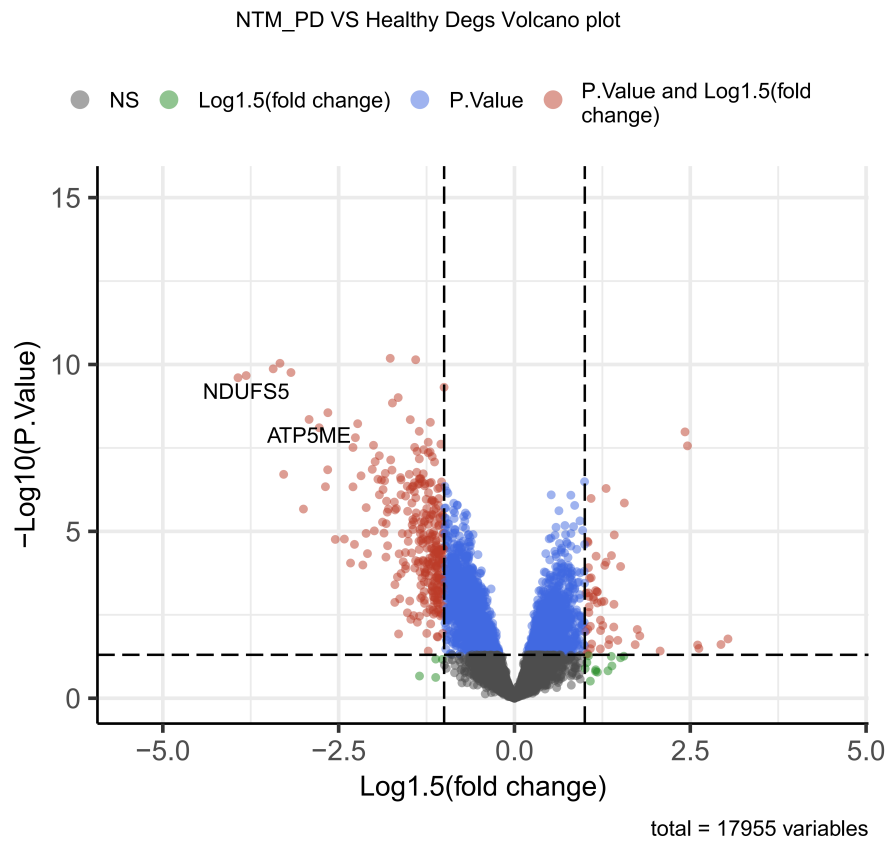


Fig. 5. This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

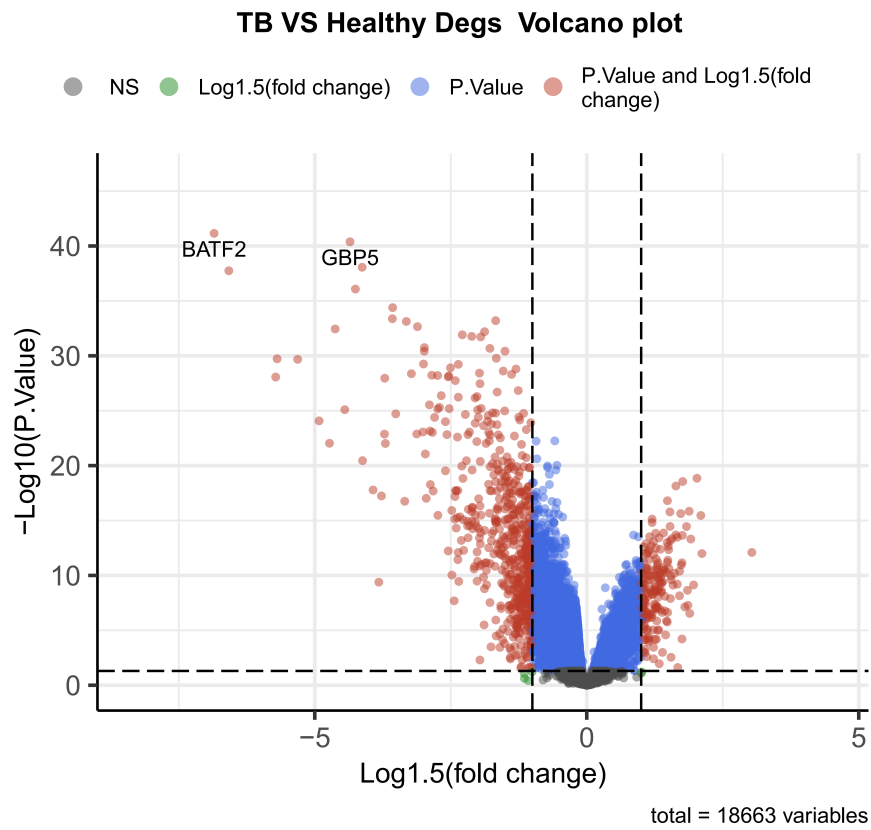


Fig. 6. This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

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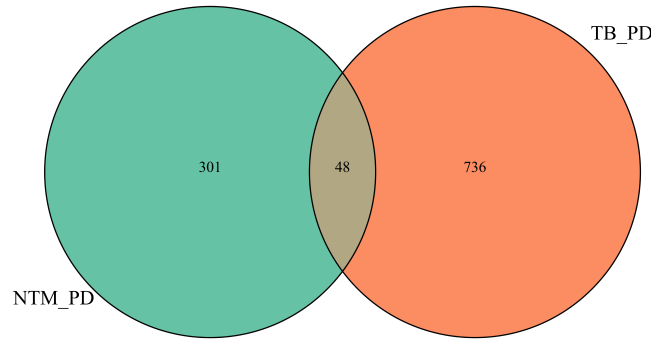


Fig. 8. This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

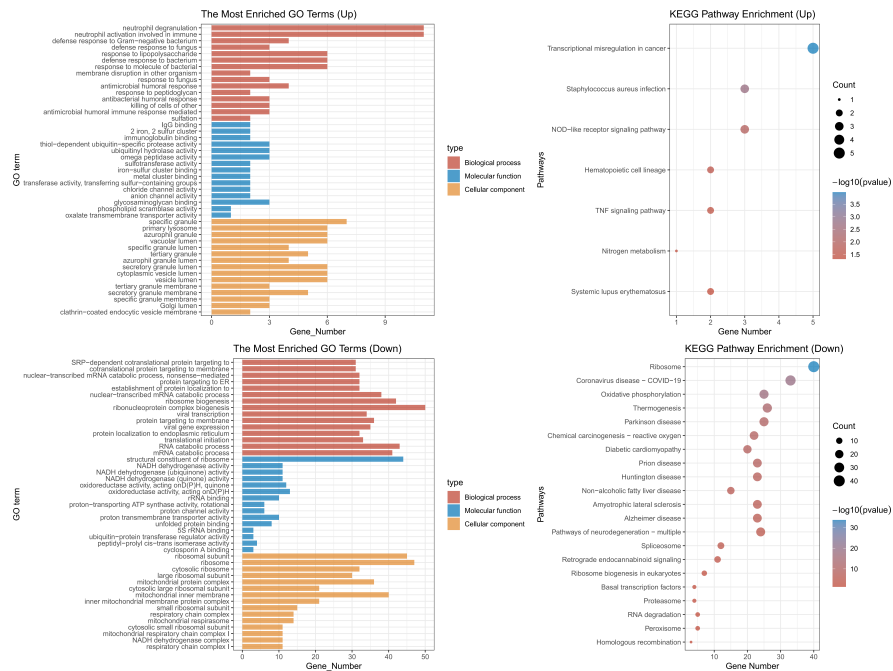


Fig. 9. This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

In case of double column layout, the above format puts figure captions/images to single column width. To get spanned images, we need to provide `\begin{figure*}` ... `\end{figure*}`.

For sample purpose, we have included the width of images in the optional argument of `\includegraphics` tag. Please ignore this.

8 Algorithms, Program codes and Listings

Packages `algorithm`, `algorithmicx` and `algpseudocode` are used for setting algorithms in L^AT_EX using the format:

```
\begin{algorithm}
\caption{<alg-caption>}\label{<alg-label>}
\begin{algorithmic}[1]
. . .
\end{algorithmic}
\end{algorithm}
```

You may refer above listed package documentations for more details before setting `algorithm` environment. For program codes, the “verbatim” package is required and the command to be used is `\begin{verbatim} ... \end{verbatim}`.

Similarly, for listings, use the `listings` package. `\begin{lstlisting} ... \end{lstlisting}` is used to set environments similar to `verbatim` environment. Refer to the `lstlisting` package documentation for more details.

A fast exponentiation procedure:

```
begin
  for i := 1 to 10 step 1 do
    expt(2,i);
    newline() od
where
proc expt(x,n) ≡
  z := 1;
  do if n = 0 then exit fi;
  do if odd(n) then exit fi;
    comment: This is a comment statement;
    n := n/2; x := x*x od;
  { n > 0 };
  n := n - 1; z := z*x od;
  print(z).
end
```

Comments will be set flush to the right margin

Algorithm 1 Calculate $y = x^n$

Require: $n \geq 0 \vee x \neq 0$ **Ensure:** $y = x^n$

```
1:  $y \leftarrow 1$ 
2: if  $n < 0$  then
3:    $X \leftarrow 1/x$ 
4:    $N \leftarrow -n$ 
5: else
6:    $X \leftarrow x$ 
7:    $N \leftarrow n$ 
8: end if
9: while  $N \neq 0$  do
10:  if  $N$  is even then
11:     $X \leftarrow X \times X$ 
12:     $N \leftarrow N/2$ 
13:  else [ $N$  is odd]
14:     $y \leftarrow y \times X$ 
15:     $N \leftarrow N - 1$ 
16:  end if
17: end while
```

```
for  $i := \text{maxint}$  to 0 do
begin
{ do nothing }
end;
Write('Case-insensitive-');
Write('Pascal-keywords-');
```

9 Cross referencing

Environments such as figure, table, equation and align can have a label declared via the `\label{#label}` command. For figures and table environments use the `\label{}` command inside or just below the `\caption{}` command. You can then use the `\ref{#label}` command to cross-reference them. As an example, consider the label declared for Figure 4 which is `\label{fig1}`. To cross-reference it, use the command `Figure \ref{fig1}`, for which it comes up as “Figure 4”.

To reference line numbers in an algorithm, consider the label declared for the line number 2 of Algorithm 1 is `\label{algl n2}`. To cross-reference it, use the command `\ref{algl n2}` for which it comes up as line 2 of Algorithm 1.

9.1 Details on reference citations

Standard $\text{\LaTeX}$ permits only numerical citations. To support both numerical and author-year citations this template uses `natbib` $\text{\LaTeX}$ package. For style guidance please refer to the template user manual.

Here is an example for `\cite{...}`: [1]. Another example for `\citep{...}`: [2]. For author-year citation mode, `\cite{...}` prints Jones et al. (1990) and `\citep{...}` prints (Jones et al., 1990).

All cited bib entries are printed at the end of this article: [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] and [13].

10 Examples for theorem like environments

For theorem like environments, we require `amsthm` package. There are three types of predefined theorem styles exists—`thmstyleone`, `thmstyletwo` and `thmstylethree`

<code>thmstyleone</code>	Numbered, theorem head in bold font and theorem text in italic style
<code>thmstyletwo</code>	Numbered, theorem head in roman font and theorem text in italic style
<code>thmstylethree</code>	Numbered, theorem head in bold font and theorem text in roman style

For mathematics journals, theorem styles can be included as shown in the following examples:

Theorem 1 (Theorem subhead) *Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proposition 2 *Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Example 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula,

eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Remark 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Definition 1 (Definition sub head) Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text.

Additionally a predefined “proof” environment is available: `\begin{proof} ... \end{proof}`. This prints a “Proof” head in italic font style and the “body text” in roman font style with an open square at the end of each proof environment.

Proof Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proof of Theorem 1 Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

For a quote environment, use `\begin{quote} ... \end{quote}`

Quoted text example. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text (refer Figure 4). Sample body text. Sample body text. Sample body text (refer Table 13).

11 Methods

Topical subheadings are allowed. Authors must ensure that their Methods section includes adequate experimental and characterization data necessary for others in the field to reproduce their work. Authors are encouraged to include RIIIDs where appropriate.

Ethical approval declarations (only required where applicable) Any article reporting experiment/s carried out on (i) live vertebrate (or higher invertebrates), (ii) humans or (iii) human samples must include an unambiguous statement within the methods section that meets the following requirements:

1. Approval: a statement which confirms that all experimental protocols were approved by a named institutional and/or licensing committee. Please identify the approving body in the methods section
2. Accordance: a statement explicitly saying that the methods were carried out in accordance with the relevant guidelines and regulations
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If your manuscript includes potentially identifying patient/participant information, or if it describes human transplantation research, or if it reports results of a clinical trial then additional information will be required. Please visit (<https://www.nature.com/nature-research/editorial-policies>) for Nature Portfolio journals, (<https://www.springer.com/gp/authors-editors/journal-author/journal-author-helpdesk/publishing-ethics/14214>) for Springer Nature journals, or (<https://www.biomedcentral.com/getpublished/editorial-policies#ethics+and+consent>) for BMC.

12 Discussion

Discussions should be brief and focused. In some disciplines use of Discussion or ‘Conclusion’ is interchangeable. It is not mandatory to use both. Some journals prefer a section ‘Results and Discussion’ followed by a section ‘Conclusion’. Please refer to Journal-level guidance for any specific requirements.

13 Conclusion

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.

In some disciplines use of Discussion or ‘Conclusion’ is interchangeable. It is not mandatory to use both. Please refer to Journal-level guidance for any specific requirements.

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

Please refer to Journal-level guidance for any specific requirements.

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Declarations

Some journals require declarations to be submitted in a standardised format. Please check the Instructions for Authors of the journal to which you are submitting to see if you need to complete this section. If yes, your manuscript must contain the following sections under the heading ‘Declarations’:

- **Funding**

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- **Conflict of Interest**

The authors declare that they have no conflict of interest.

- **Ethics approval and consent to participate**

This study was approved by the Ethics Review Committee of the First Affiliated Hospital of Guangxi Medical University (2020.KY-E-100). Written informed consent was provided by all participants in the prospective cohort study. All patients were followed-up until July 1, 2020, or until the date of death. All methods were carried out in accordance with the Declaration of Helsinki guidelines and regulations.

- **Consent for publication**

Written informed consent for publication was obtained from all individual participants included in the study.

- **Data availability**

The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

- **Abbreviations**

TB: Tuberculosis

NTM: Non-tuberculosis mycobacterium

BMI: Body mass index

AUC: Area under the curve

ROC: Receiver operating characteristic

HIV: Human immunodeficiency virus

- **Author contributions**

C.L. (Liang Chaoyue) designed the study, performed data analysis, and drafted the main manuscript as the first author. H.L. (Luo Hongyun), C.Z. (Zhou Chuanlin), Q.T. (Tang Qiulian), X.Y. (Yu Xiuwen), R.X. (Xie Rui), Y.X. (Xu Yang), and C.Zh. (Zhang Chuyue) contributed to data collection, interpretation, and manuscript revision. J.L. (Luo Jin) and J.K. (Kong Jinliang) supervised the project and are the corresponding authors. All authors reviewed and approved the final manuscript.

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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